

價值投資選股策略 | DCF 現金流折現法

DCF 通常在哪些科系學到？

- 財務金融學系 / 金融工程學系
屬於核心課程內容，因為估值是金融學的基石。
- 會計系（特別是管理會計、財務會計延伸課程）
會在「公司價值評估」或「財務管理」課程中提到。
- 企業管理學系（財務管理方向）
作為投資決策與企業併購的工具之一。
- MBA、EMBA
幾乎一定會教，特別是在「公司財務 (Corporate Finance)」與「投資學 (Investments)」課程。

現金流折現法 (DCF, Discounted Cash Flow)

$$DCF = \sum_{t=1}^n \frac{FCF_t}{(1 + WACC)^t}$$

- FCF_t = 第t年的自由現金流 (Free Cash Flow)
- WACC = 加權平均資本成本 (Weighted Average Cost of Capital)
- t = 年份 (1, 2, 3, ... n)
- Σ = 加總符號 (從第1年到第n年)

註：在企業估值中，折現率通常使用WACC

加權平均資本成本

$$WACC = \frac{E}{V} \cdot R_e + \frac{D}{V} \cdot R_d \cdot (1 - T)$$

- E = 股權市值 (Market Value of Equity)
- D = 債務市值 (Market Value of Debt)
- $V = E + D$ = 企業總價值 (Total Value)
- R_e = 股權成本 (Cost of Equity)
- R_d = 債務成本 (Cost of Debt)
- T = 有效稅率 (Tax Rate)

股權成本(CAPM模型)

$$R_e = R_f + \beta \cdot (R_m - R_f)$$

- R_f = 無風險利率 (Risk-free Rate)
- β = Beta係數 (系統風險係數)
- R_m = 市場預期報酬率
- $(R_m - R_f)$ = 市場風險溢價 (Market Risk Premium)

企業價值 (Enterprise Value, EV)

$$EV = \sum_{t=1}^n \frac{FCF_t}{(1 + WACC)^t} + \frac{TV}{(1 + WACC)^n}$$

- EV = 企業價值 (Enterprise Value)
- PV = 現值 (Present Value)
- FCF_t = 第t年自由現金流
- WACC = 加權平均資本成本
- TV = 終值 (Terminal Value)
- n = 明確預測期年數

$$EV = \sum_{t=1}^n \frac{FCF_t}{(1 + WACC)^t} + \frac{FCF_{n+1}}{WACC - g} \cdot \frac{1}{(1 + WACC)^n}$$

終值(Terminal Value)

$$TV = \frac{FCF_{n+1}}{WACC - g}$$

- g 是永續增長率

每股價值計算

$$P_0 = \frac{EV - NetDebt + Cash}{N}$$

每股價值 = (EV - 淨債務 + 現金) ÷ 流通股數

Share Price = (EV - Net Debt + Cash) ÷ Shares Outstanding

- EV = 企業價值
- Net Debt : 淨債務 = 總債務 - 現金及約當現金
- Cash = 現金及短期投資
- N(Shares Outstanding)= 流通股數

FMP DCF API

Endpoint:

https://financialmodelingprep.com/stable/custom-discounted-cash-flow?symbol=AAPL

Parameters

Query Parameter	Type	Example
symbol *	string	AAPL
revenueGrowthPct	number	0.1094119804597946
ebitdaPct	number	0.31273548388
depreciationAndAmortizationPct	number	0.0345531631720999
cashAndShortTermInvestmentsPct	number	0.2344222126801843
receivablesPct	number	0.1533770531229388
inventoriesPct	number	0.0155245674227653
payablePct	number	0.1614868903169657
ebitPct	number	0.2781823207138459
capitalExpenditurePct	number	0.0306025847141713
operatingCashFlowPct	number	0.2886333485760204
sellingGeneralAndAdministrativeExpensesPct	number	0.0662854095187211
taxRate	number	0.14919579658453103
longTermGrowthRate	number	4
costOfDebt	number	3.64
costOfEquity	number	9.51168
marketRiskPremium	number	4.72
beta	number	1.244
riskFreeRate	number	3.64

https://financialmodelingprep.com/stable/custom-discounted-cash-flow?symbol=AAPL&apikey=

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48    "equityValuePerShare": 195.61,
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